Explainable Cross-Attention Multimodal Framework for Breast Cancer Diagnosis

OBJECTIVES OF RESEARCH

- O1: To design and develop a multimodal deep learning framework that effectively fuses **mammography images** with **structured clinical metadata** (breast density, mass shape, margins, and calcification details) to improve diagnostic accuracy beyond single-modality approaches.
- O2: To implement a specialized preprocessing pipeline that transforms raw **textual clinical labels from CSV datasets** (specifically the CBIS-DDSM mass and calcification descriptions) into high-dimensional one-hot encoded vectors suitable for neural network integration.
- O3: To develop a **bidirectional cross-attention mechanism** that enables the model to dynamically "attend" to specific image regions based on the corresponding clinical findings, facilitating a synergistic interaction between visual and categorical evidence.
- O4: To enhance model robustness through **modality dropout and imputation strategies**, ensuring the framework remains functional and accurate even when certain clinical features are missing from the patient record.
- O5: To provide clinical transparency by integrating **Explainable AI (XAI) tools**, including SHAP and Grad-CAM, to visualize which specific clinical features and image pixels most heavily influenced the model's final diagnostic decision.
- O6: To validate the proposed system's performance against state-of-the-art benchmarks using the **CBIS-DDSM and TCGA datasets**, focusing on metrics of sensitivity, specificity, and the Area Under the Curve (AUC).
- O7: To design and develop a real-time diagnostic framework using **Flask**, providing a functional web-based interface where clinicians can upload mammogram images and input clinical features via a structured form to receive immediate, explainable diagnostic support.

PROBLEM STATEMENT

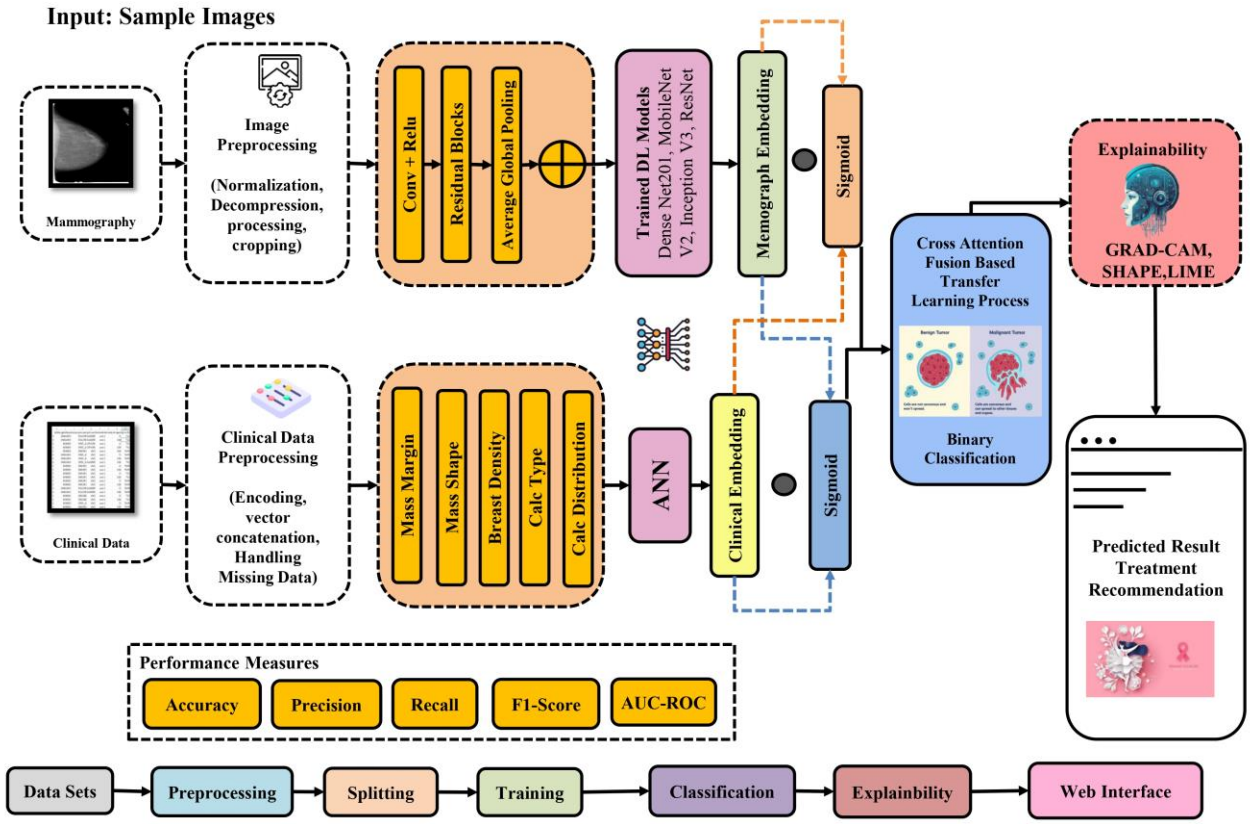
Breast cancer remains a leading cause of mortality among women worldwide, requiring early and highly accurate diagnosis for effective treatment. Current Computer-Aided Diagnosis (CAD) systems predominantly rely on mammography images alone, often overlooking the critical diagnostic evidence contained in clinical reports, such as breast density, mass shape, and margins. This unimodal approach limits the model's ability to capture the holistic "patient picture," leading to lower sensitivity or false positives.

Furthermore, most existing multimodal models act as "black boxes," providing predictions without explaining *why* a specific diagnosis was reached, which hinders clinical trust. There is also a significant gap between theoretical AI models and clinical practice; many high-performing models lack a functional, user-friendly interface for real-time application.

This research addresses these challenges by:

1. **Data Integration Gap:** Developing a mechanism to transform raw, textual clinical metadata from standard medical datasets (like CBIS-DDSM) into structured numerical features for deep learning.
2. **Fusion Complexity:** Utilizing a bidirectional cross-attention mechanism to model the complex relationship between visual mammographic features and categorical clinical data.
3. **Lack of Transparency:** Incorporating Explainable AI (XAI) techniques (Grad-CAM and SHAP) to justify decisions to medical professionals.
4. **Accessibility & Real-time Utility:** Closing the gap between research and practice by designing a **Flask-based web framework** that allows for the real-time upload and processing of mammograms and clinical features for immediate diagnostic support.

PROPESED RESEARCH METHODOLOGY



DATASET

CBIS-DDSM Memmogram Images:

i. 1592 Mass Images
ii. 1511 Calcification Images

DICOM Images change into png by using TCIA Data Reteriver

CBIS-DDSM Clinical Data .csv Files:

Data Access

Version 1: Updated 2017/09/14

Title	Data Type	Format	Access Points	Subjects	Studies	Series	Images	License	Metadata
Images	MG	DICOM	DOWNLOAD (163.51GB) SEARCH Download requires TCIA Data Retriever	1,566	2,620	6,775	10,239	CC BY 3.0	View
Mass-Training-Description	Classification	CSV	DOWNLOAD (754.62KB)	691				CC BY 3.0	—
Calc-Training-Description	Classification	CSV	DOWNLOAD (903.62KB)	602				CC BY 3.0	—
Mass-Test-Description	Classification	CSV	DOWNLOAD (211.95KB)	201				CC BY 3.0	—
Calc-Test-Description	Classification, Radiomic Feature	CSV	DOWNLOAD (185.93KB)	151				CC BY 3.0	—